EXCELLENCE

CLASS 11 - 2026-27

Physics | Units 04

Units and Measurement

Significant Figures, Accuracy and Errors

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Topics to be covered

01 Significant figures in a calculated answer

02 Rounding off

03 Accuracy and precision

04 Absolute, relative and percentage error

05 Mean error from repeated readings

06 Dimensional questions with a new set of quantities

07 When a formula cannot be built by dimensions

Where we left Units 03

Last class we used dimensions to find powers, and we learned how to count significant digits in one measured number. A trailing zero after the decimal counts. A leading zero does not. The power of ten is not a digit.

Today we use those digits in a sum, a difference, a product and a quotient. Then we separate accuracy from precision, and we start errors: absolute, relative, percentage, and the mean of repeated readings.

A calculated answer keeps only the digits the measurements can support.

Class notes will be saved after teaching. This deck is for the board.

What a significant figure is

A significant figure is a digit we trust in a measurement. All the certain digits are kept, and one uncertain digit is kept with them. That last digit is the one the instrument is guessing between two marks.

Example: a mass written as 256703 kg has six significant digits. Five of them are certain. The last one, 3, is the uncertain digit. More significant digits means a finer measurement. That is precision.

Significant figures = all certain digits + one uncertain digit

Changing the unit does not change the count

2.5 kg and 2.5×10^3 g are the same mass. Both have two significant digits. Writing 2500 g would look like a different count, so we keep scientific notation when we change the unit.

2.5 kg = 2.5×10^3 g both have 2 significant digits

The power 10^3 is only a place-value label. It is not an extra digit.

Rough work / extra work

How to count — part 1

- Every non-zero digit is significant. 2427 m has 4.
- A zero between non-zero digits is significant. 24507 has 5.
- A trailing zero with no decimal point is not significant. 4500 m has 2.
- Zeros after the decimal point are significant. 45.00 cm has 4.

4500 m and 45.00 cm are not the same kind of zero.

How to count — part 2

- In a number smaller than 1, zeros on the left of the first non-zero digit are not significant. 0.02 m has 1. 0.0307 m has 3.

- An exact count has unlimited significant digits. Five pens means exactly 5, not a measured 5.

- In scientific notation, only the front number is counted. 2.303×10^{23} m has 4 significant digits.

Exact number → unlimited digits. Power of ten → not a digit.

Count these measurements

How many significant figures are in each?

1. 36.7 m
2. 0.006606 s
3. 2, 002 kg
4. 306, 490, 000 people, counted exactly

Solution: Count these measurements

Apply one rule to each number.

Step 1: 36.7 m — three non-zero digits → 3.

Step 2: 0.006606 s — leading zeros drop. Digits 6, 6, 0, 6 → 4.

Step 3: 2, 002 kg — zeros sit between non-zeros → 4.

Step 4: An exact people-count is not a measured guess → unlimited significant figures.

3 , 4 , 4 , unlimited

Three numbers together

P = 0.0030 m, Q = 2.40 m, R = 3000 m.

Significant figures in P, Q, R are

1. 1, 2, 1
2. 2, 3, 1
3. 4, 2, 1
4. 4, 2, 4

Solution: Three numbers together

Read each number with its own zero rule.

Step 1: P = 0.0030. Leading zeros drop. The last 0 is after the decimal → 2.

Step 2: Q = 2.40. The zero after the decimal counts → 3.

Step 3: R = 3000. No decimal point, so trailing zeros drop → 1.

Final answer: 2, 3, 1 (option 2)

Rough work / extra work

Addition and subtraction

When you add or subtract, the answer is limited by the least precise decimal place. The calculator may print extra digits. Those extra digits are not in the measurements, so they are cut off.

4.71 + 1.3 = 6.01 on a calculator. 1.3 has one decimal place, so the sum is 6.0.

3.813 + 2.21 = 6.023. 2.21 has two decimal places, so the sum is 6.02.

Sum or difference → keep the least number of decimal places.

Multiplication and division

When you multiply or divide, the answer keeps the smaller count of significant figures. Decimal places are not the rule here. The count of trusted digits is the rule.

3.01 × 3.1 = 9.331 on a calculator.

3.01 has 3 significant figures. 3.1 has 2. Keep 2 → 9.3.

Product or quotient → keep the least number of significant figures.

Area of a rectangle

Length 55.3 m, breadth 25 m. The area, with the proper significant figures, is

1. $1382\ m^2$
2. $1382.5\ m^2$
3. $14 \times 10^2\ m^2$
4. $138 \times 10^1\ m^2$

Solution: Area of a rectangle

Area is a product, so count significant figures, not decimal places.

Step 1: $A = 55.3 \times 25 = 1382.5\ m^2$ on the calculator.

Step 2: 55.3 has 3 significant figures. 25 has 2. Keep 2.

Step 3: 1382.5 to 2 significant figures is 1.4×10^3, which is the same as 14×10^2.

Final answer: $14 \times 10^2\ m^2$ (option 3)

Rough work / extra work

A small subtraction

With significant figures, the value of 9.99 m – 0.0099 m is

1. 9.98 m
2. 9.980 m
3. 9.9 m
4. 9.9801 m

Solution: A small subtraction

This is a difference, so match decimal places.

Step 1: 9.99 − 0.0099 = 9.9801 on the calculator.

Step 2: 9.99 has two decimal places. 0.0099 has four. Keep two.

Step 3: 9.9801 written to two decimal places is 9.98 m.

Final answer: 9.98 m (option 1)

Add three lengths

Add 0.307 m, 0.52 m and 0.4 m.

1. 1.227 m
2. 1.2 m
3. 1.3 m
4. 1.23 m

Solution: Add three lengths

The roughest length decides the decimal place.

Step 1: 0.307 + 0.52 + 0.4 = 1.227 m.

Step 2: Decimal places available: 3, 2 and 1. The least is 1, from 0.4 m.

Step 3: 1.227 to one decimal place is 1.2 m.

Final answer: 1.2 m (option 2)

The three rounding rules

- If the first digit you drop is less than 5, leave the last kept digit as it is.
- If the first digit you drop is more than 5, increase the last kept digit by 1.
- If the first digit you drop is exactly 5, leave an even last digit unchanged, and increase an odd last digit by 1.

…4 stays. …6 goes up. …5 goes to the even digit.

Rough work / extra work

Round to 4 significant figures

(i) Round 4.7475 to 4 significant figures.

(ii) Round 4.7465 to 4 significant figures.

Solution: Round to 4 significant figures

Four significant figures means we keep 4.74x and look at the next digit, which is 5.

Step 1: 4.7475 — the kept digit before the 5 is 7, which is odd, so it becomes 8.

Step 2: 4.7465 — the kept digit before the 5 is 6, which is even, so it stays 6.

(i) 4.748 (ii) 4.746

Proper final answers

(i) 101.2 + 18.702 = ?

(ii) 202.88 – 1.013 = ?

Solution: Proper final answers

Both are add or subtract, so use decimal places.

Step 1: 101.2 + 18.702 = 119.902. Least decimal places = 1 → 119.9.

Step 2: 202.88 − 1.013 = 201.867. Least decimal places = 2 → 201.87.

(i) 119.9 (ii) 201.87

One sum and one quotient

(i) A = 2.413 and B = 1.2. Find A + B.

(ii) A = 25.5 and B = 5, both measured. Find A/B.

Solution: One sum and one quotient

The sum follows decimal places. The quotient follows significant figures.

Step 1: 2.413 + 1.2 = 3.613. B has one decimal place → 3.6.

Step 2: 25.5 / 5 = 5.1 on the calculator. Measured 5 has 1 significant figure.

Step 3: So A/B is written as 5. If the 5 were an exact count, we would keep 5.10.

(i) 3.6 (ii) 5

Rough work / extra work

Accuracy and precision

Accuracy tells how close a reading is to the true value. A small error from the true value means a more accurate reading.

Precision tells how fine the instrument is, and how close repeated readings are to each other. More decimal places, or more significant figures, means a more precise reading. A precise reading can still sit far from the true value.

Accuracy → close to the true value. **Precision → a fine, repeatable reading.**

Most accurate, most precise

True length = 6.57 m. Which reading is most accurate, and which is most precise?

1. 6.52 m
2. 6.61 m
3. 6.513 m
4. 6.68 m

Solution: Most accurate, most precise

Accuracy uses the gap from 6.57 m. Precision uses how fine the number is.

Step 1: Gaps: 6.52 → 0.05, 6.61 → 0.04, 6.513 → 0.057, 6.68 → 0.11.

Step 2: Smallest gap is 0.04, so 6.61 m is the most accurate.

Step 3: 6.513 m is written to a thousandth of a metre, so it is the most precise.

Most accurate: 6.61 m. Most precise: 6.513 m.

Finest reading

No true value is given. Which reading is the finest?

1. 2.4 m
2. 2.41 m
3. 2 m
4. 2.413 m

Solution: Finest reading

Accuracy needs a true value. Here we only have the readings, so we judge precision.

Step 1: 2 m has 1 decimal-free digit. 2.4 m has 2 significant figures.

Step 2: 2.41 m has 3. 2.413 m has 4, down to a millimetre.

Most precise reading: 2.413 m (option 4)

Do not call it more accurate until a true value is given.

Rough work / extra work

Three ways to write an error

Absolute error is the size of the mistake, in the same unit as the quantity. If the true value is a_0 and the measured value is a, the absolute error is $|a - a_0|$.

Relative error is that absolute error divided by the true value. It has no unit. Percentage error is the relative error multiplied by 100.

Relative error = absolute error / true value. **Percentage error = relative error × 100.**

Relative error

Absolute error = 5 and true value = 15. The relative error is

Solution: Relative error

Relative error is a ratio, so the unit cancels.

Step 1: Relative error = 5 / 15 = 1/3.

Step 2: Percentage error = (1/3) × 100 = 33.3%.

Relative error = 1/3

Error in a sum

$l_1 = (10 \pm 2)$ cm and $l_2 = (20 \pm 1)$ cm. Find $l = l_1 + l_2$ with its error.

Solution: Error in a sum

For a sum, the absolute errors are added. They are not averaged.

Step 1: $l = 10 + 20 = 30$ cm.

Step 2: $\Delta l = 2 + 1 = 3$ cm.

$l = (30 \pm 3)$ cm

Rough work / extra work

Mean and mean absolute error

When the same quantity is measured several times, the mean is the best single value. Each reading then has its own absolute error, equal to how far it sits from the mean.

The mean absolute error is the average of those gaps. The result is written as mean ± mean absolute error. Relative error is that mean absolute error divided by the mean.

Result = mean ± mean absolute error

Pendulum period

Five readings of the time period, in seconds:

2.63, 2.56, 2.42, 2.71, 2.80

Find the mean, the mean absolute error, the relative error and the percentage error.

Solution: mean of the period

Add the five readings, then divide by 5.

Step 1: Sum = 2.63 + 2.56 + 2.42 + 2.71 + 2.80 = 13.12 s.

Step 2: Mean T = 13.12 / 5 = 2.624 s.

Step 3: We will report this as 2.62 s once the error is known to the hundredth.

Solution: errors in the period

Gap of each reading from 2.624 s.

Step 1: 0.006, 0.064, 0.204, 0.086, 0.176. Sum of gaps = 0.536 s.

Step 2: Mean absolute error = 0.536 / 5 = 0.107 s ≈ 0.11 s.

Step 3: Relative error = 0.11 / 2.62 ≈ 0.04. Percentage error ≈ 4%.

T = (2.62 ± 0.11) s, relative error 0.04, percentage error 4%

Rough work / extra work

Matching powers still works

Write the unknown as a product of powers. Put dimensions on both sides. Equate the powers of M, of L and of T separately. A pure number such as 2π or 6π never comes out of this method.

If two quantities in the list have the same dimensions, they cannot both be used as separate base quantities. If a quantity is an angle, it is a pure number and cannot build a force or a time by itself.

Same dimensions, or a pure angle, means the method stops.

Electric field and magnetic field

The relation between [E] and [B] is

1. $[E] = [B]\,[L]\,[T]$
2. $[E] = [B]\,[L]^{-1}\,[T]$
3. $[E] = [B]\,[L]\,[T]^{-1}$
4. $[E] = [B]\,[L]^{-1}\,[T]^{-1}$

Solution: Electric field and magnetic field

In an electromagnetic wave, E = c B. Speed c carries the extra dimensions.

Step 1: $[c] = L\,T^{-1}$, so $[E] = [B] \times L\,T^{-1}$.

Step 2: That is $[E] = [B]\,[L]\,[T]^{-1}$.

Final answer: option 3

Permittivity and permeability

The relation between [ε_0] and [μ_0] is

1. $[\mu_0] = [\varepsilon_0] [L]^2 [T]^{-2}$
2. $[\mu_0] = [\varepsilon_0] [L]^{-2} [T]^2$
3. $[\mu_0] = [\varepsilon_0]^{-1} [L]^2 [T]^{-2}$
4. $[\mu_0] = [\varepsilon_0]^{-1} [L]^{-2} [T]^2$

Solution: Permittivity and permeability

Use $c = 1 / \sqrt{(\mu_0 \varepsilon_0)}$. Square both sides.

Step 1: $\mu_0 \varepsilon_0 = 1 / c^2$, so $[\mu_0]\,[\varepsilon_0] = L^{-2} T^2$.

Step 2: $[\mu_0] = [\varepsilon_0]^{-1} [L]^{-2} [T]^2$.

Final answer: option 4

Rough work / extra work

Same dimensions as S

S = (E × B) / μ_0. Its dimensions match which of these?

1. Energy / (charge × current)
2. Force / (length × time)
3. Energy / volume
4. Power / area

Solution: Same dimensions as S

Build [S] from the known formulas, then compare.

Step 1: $[E] = M\,L\,T^{-3}\,I^{-1}$, $[B] = M\,T^{-2}\,I^{-1}$, $[\mu_0] = M\,L\,T^{-2}\,I^{-2}$.

Step 2: $[E\,B / \mu_0] = M\,T^{-3}$.

Step 3: Force / (length × time) = $M\,T^{-3}$. Power / area = $M\,T^{-3}$.

Final answer: options 2 and 4

Mass and angular momentum are pure numbers

In a new system, mass and angular momentum are dimensionless, and length has dimension L. Which statements are correct?

1. Dimension of force is L^{-3}
2. Dimension of energy is L^{-2}
3. Dimension of power is L^{-5}
4. Dimension of linear momentum is L^{-1}

Solution: Mass and angular momentum are pure numbers

Angular momentum m v r is a pure number, and m is a pure number, so velocity must cancel the length.

Step 1: [v] = L^{-1}. Then [time] = length / speed = L^{2}.

Step 2: [force] = [mass × acceleration] = L^{-3}. [energy] = force × length = L^{-2}.

Step 3: [power] = energy / time = L^{-4}, not L^{-5}. [momentum] = m v = L^{-1}.

Correct: options 1, 2 and 4. Option 3 is wrong.

Young's modulus in F, A and V

Speed V, area A and force F are the base quantities. [Young's modulus] is

1. $F\,A^{-1}\,V^{0}$
2. $F\,A^{2}\,V^{-1}$
3. $F\,A^{2}\,V^{-2}$
4. $F\,A^{2}\,V^{-3}$

Solution: Young’s modulus in F, A and V

Young’s modulus is stress / strain, and strain is a pure number.

Step 1: Y = F / A.

Step 2: Speed is not needed, so its power is 0.

Final answer: $F\ A^{-1}\ V^{0}$ (option 1)

Rough work / extra work

Density from force, velocity and time

F, V and T are base quantities. The dimensional formula of density is

1. $F\ V^{4}\ T^{-6}$
2. $F\ V^{-4}\ T^{-2}$
3. $F^{2}\ V^{-2}\ T^{6}$
4. $F\ V^{-2}\ T^{2}$

Solution: Density from force, velocity and time

Assume density = $F^a\ V^b\ T^c$ and match M, L, T.

Step 1: Mass: a = 1. Length: a + b = -3, so b = -4.

Step 2: Time: -2a - b + c = 0 → -2 - (-4) + c = 0 → c = -2.

Final answer: $F\ V^{-4}\ T^{-2}$ (option 2)

Viscosity from momentum, area and time

P, A and T are base quantities. The dimensional formula of viscosity is

1. $[P\ A^{-1}]$
2. $[P\ A\ T^{-1}]$
3. $[P\ A^{-1}\ T]$
4. $[P\ A^{-1}\ T^{-1}]$

Solution: Viscosity from momentum, area and time

From $F = 6\pi \eta r v$, the dimensions of η are $M L^{-1} T^{-1}$.

Step 1: $\eta = P^a A^b T^c$. Mass gives a = 1.

Step 2: Length: $1 + 2b = -1$, so $b = -1$. Time: $-1 + c = -1$, so $c = 0$.

Final answer: $[P A^{-1}]$ (option 1)

Rough work / extra work

Mass from time, velocity and angular momentum

Express mass using time t, velocity v and angular momentum J.

Solution: Mass from time, velocity and angular momentum

[J] = M L^2 T^{-1}. Assume M = J^a v^b t^c.

Step 1: Mass: a = 1. Length: 2 + b = 0, so b = -2.

Step 2: Time: -1 - (-2) + c = 0, so c = -1.

Mass = J / (v^2 t)

Young's modulus from c, h and G

Young's modulus is a product of powers of c, h and G. Which powers are correct?

1. $\alpha = 7, \beta = -1, \gamma = -2$
2. $\alpha = -7, \beta = -1, \gamma = -2$
3. $\alpha = 7, \beta = -1, \gamma = 2$
4. $\alpha = -7, \beta = 1, \gamma = -2$

Solution: equations for Y

$[Y] = M\,L^{-1}\,T^{-2}$, $[c] = L\,T^{-1}$, $[h] = M\,L^{2}\,T^{-1}$, $[G] = M^{-1}\,L^{3}\,T^{-2}$.

Step 1: Mass: $\beta - \gamma = 1$.

Step 2: Length: $\alpha + 2\beta + 3\gamma = -1$.

Step 3: Time: $-\alpha - \beta - 2\gamma = -2$, so $\alpha + \beta + 2\gamma = 2$.

Solution: the three powers

From mass, $\beta = 1 + \gamma$. Put that into the time equation.

Step 1: $\alpha + (1 + \gamma) + 2\gamma = 2 \rightarrow \alpha + 3\gamma = 1$.

Step 2: Length then gives $\alpha + 5\gamma = -3$. Subtract: $2\gamma = -4$, so $\gamma = -2$.

Step 3: $\alpha = 1 - 3(-2) = 7$. $\beta = 1 + (-2) = -1$.

$\alpha = 7, \beta = -1, \gamma = -2$ (option 1)

Rough work / extra work

A leaking balloon

Gas speed depends on the radius through a power a. The emptying time T depends on surface tension S, outlet area A, density ρ and starting radius R through powers α, β, γ and δ.

1. $a = 1/2, \alpha = 1/2, \beta = -1, \gamma = 1, \delta = 3/2$
2. $a = -1/2, \alpha = -1/2, \beta = -1, \gamma = -1/2, \delta = 5/2$
3. $a = -1/2, \alpha = -1/2, \beta = -1, \gamma = 1/2, \delta = 7/2$
4. $a = 1/2, \alpha = 1/2, \beta = -1/2, \gamma = 1/2, \delta = 7/2$

Solution: speed of the gas

The speed of the leaving gas is fixed by S, ρ and the present radius r.

Step 1: $v \propto S^{1/2} \rho^{-1/2} r^{-1/2}$.

Step 2: Mass cancels. What remains is a speed, and the power of r is -1/2. So a = -1/2.

Solution: time to empty

Volume leaves through area A, so the radius falls as $dr/dt \propto A\, v / r^2$.

Step 1: With $v \propto r^{-1/2}$, we get $dr/dt \propto A\, S^{1/2}\, \rho^{-1/2}\, r^{-5/2}$.

Step 2: Time to fall from R to 0 is $T \propto R^{7/2} / (A\, S^{1/2}\, \rho^{-1/2})$.

Step 3: So $\alpha = -1/2$, $\beta = -1$, $\gamma = 1/2$, $\delta = 7/2$.

Final answer: option 3

Three cases where dimensions refuse

- Torque, energy and work have the same dimensions. Using all three as base quantities cannot produce a new formula for time.

- If g is said to depend on force, energy, mass and velocity, that is four quantities and only three equations (M, L, T). The powers are not unique.

- If force is said to depend on an angle, a velocity and an energy, the angle is a pure number. It cannot supply the dimensions of force.

Write “not possible” and name the reason. That is a full answer.

Which equation cannot be fully derived

Which equation cannot be obtained from dimensions alone?

1. $F = 6\pi \eta r v$
2. $\theta = \omega t$
3. $P = \rho g h$
4. $v = d / t$

Solution: Which equation cannot be fully derived

Dimensions can give the powers. They cannot give a pure number such as 6π.

Step 1: F, η, r and v can be matched, but 6π stays unknown.

Step 2: θ = ω t, P = ρ g h and v = d / t are products of powers. Their number in front is 1.

Final answer: F = 6π η r v (option 1)

Rough work / extra work

Units 04 lock lines

Add or subtract → least decimal places. Multiply or divide → least significant figures.

A dropped 5 rounds the last kept digit to the even choice.

Accuracy is closeness to the true value. Precision is how fine the reading is.

Relative error = absolute error / true value. For a sum, add the absolute errors.

Repeated readings: report mean ± mean absolute error.

Ready theory lines

1. Significant figures are the certain digits plus one uncertain digit.

2. Changing the unit does not change the number of significant figures.

3. In addition and subtraction the result has the least number of decimal places.

4. In multiplication and division the result has the least number of significant figures.

5. Absolute error has a unit. Relative error and percentage error do not add a new kind of unit.

Percentage error = (absolute error / true value) × 100.

Practice

1. Round 2.745 and 2.735 to three significant figures.

2. A = 12.5 cm, B = 0.30 cm. Find A × B and A − B with the proper digits.

3. True value 9.80. Readings 9.79, 9.85, 9.77. Which is most accurate?

4. x = (5.0 ± 0.2) and y = (3.0 ± 0.1). Find x + y with the error.

5. If resistance R depends on V and I only, show why a pure number in R = k V/I cannot be found by dimensions.

Next class: how errors combine in a product, a quotient and a power.